

Clinical Data Visualizer – User Guide

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1 Introduction

Clinical Data Visualizer is an interactive dashboard for visualizing clinical physiological signals. It allows clinicians and researchers to explore, compare, and annotate time-series data from multiple medical devices in a single unified interface.

1.1 Key Features

- **Multi-source visualization:** Display signals from up to 9 different clinical data sources simultaneously.
- **Interactive plots:** Zoom, pan, and explore data at any time scale with automatic resampling.
- **Annotations:** Draw lines and rectangles directly on plots to mark events or regions of interest. Annotations are saved and persist across sessions.
- **Flexible configuration:** Choose which signals to display, customize labels, units, colors, and group related signals together.
- **Export:** Generate standalone HTML visualizations for sharing.

1.2 Supported Data Sources

Data Source	Description
Philips Waves	High-frequency waveform data from Philips patient monitors
Philips Numerics	Numeric parameter data from Philips monitors (heart rate, SpO2, etc.)
EIT (PulmoVista)	Electrical Impedance Tomography data from Draeger PulmoVista
FluxMed Signals	Waveform data from FluxMed respiratory monitors
FluxMed Parameters	Parameter data from FluxMed monitors
Servo-U	Ventilator data from Getinge Servo-U
Mindray	Patient monitor data from Mindray scopes
Syringe	Syringe pump infusion data
Other (Generic)	Auto-discovers CSV or Parquet files with datetime columns

2 Launching the Application

2.1 Starting the App

Locate the **ClinicalVisuAppAlexis** executable in the application folder and double-click it.

A terminal window will appear showing the application starting up. After a few seconds, your default web browser will automatically open at:

`http://127.0.0.1:8050`

If the browser does not open automatically, manually navigate to the address above.

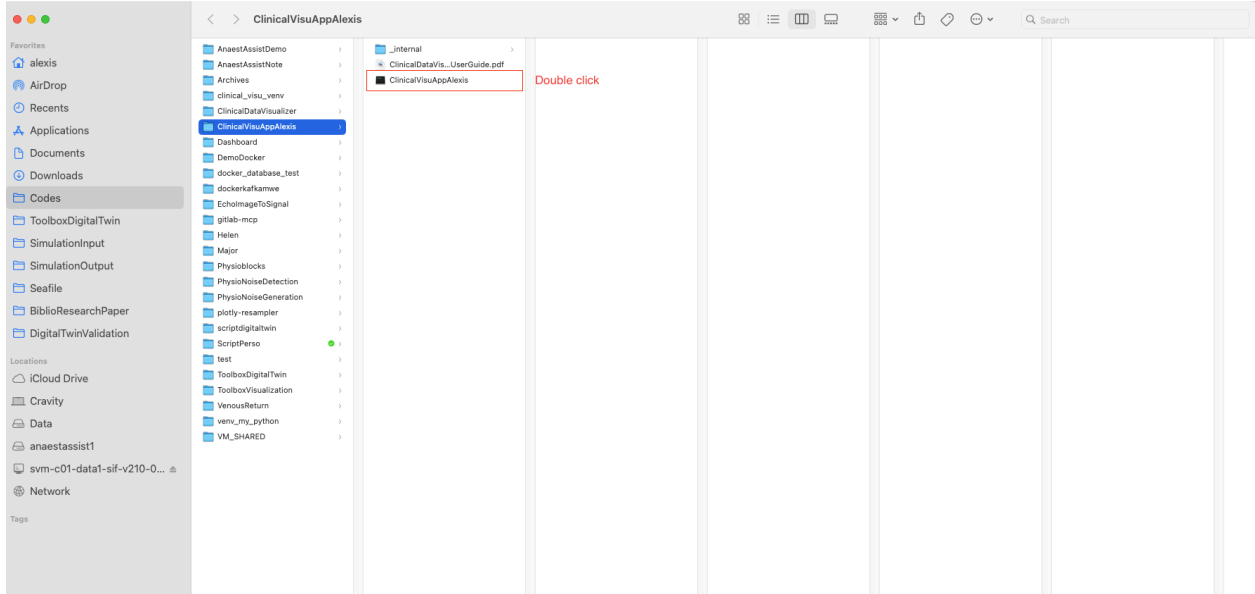


Figure 1: Application launch screen

2.2 Application Overview

The interface is organized top-to-bottom in the following order:

1. **Database Options** – Load or select a visualization configuration.
2. **Patient Options** – Configure data folder, time range, and per-source settings.
3. **Process Button** – Start the visualization.
4. **Annotations Controls** – Manage annotations (visible after processing).
5. **Visualization Area** – Interactive plots.

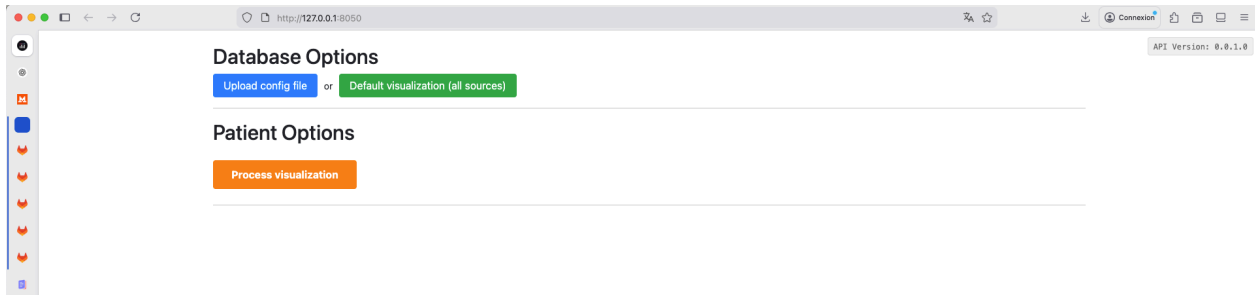


Figure 2: Application main interface

3 Preparing Your Data

3.1 Patient Folder Structure

Each patient's data must be organized in a root folder with **one subfolder per data source**. The application automatically identifies data sources based on keywords in subfolder names.

```
Patient1/
  philips_waves/      Philips waveform data (.parquet)
  philips_numerics/   Philips numeric data
  eit/               EIT PulmoVista data (.asc)
  fluxmed_signals/    FluxMed waveform data
  fluxmed_parameters/ FluxMed parameter data
  servo_u/           Servo-U ventilator data (.sta)
  mindray/           Mindray scope data (.xml or .csv)
  syringe/           Syringe pump data
  other/             Generic data (.csv or .parquet)
  tdv_visu/          Auto-created: cached data and outputs
```

You only need subfolders for the data sources you actually have. Empty or missing subfolders are simply skipped.

3.2 Folder Naming Rules

Folder names are **flexible** – they just need to contain the required keywords (case-insensitive, any separator allowed):

Data Source	Required Keywords	Examples
Philips Waves	philips + waves	philips_waves, Philips-Waves, waves_philips
Philips Numerics	philips + numerics	philips_numerics, Philips-Numerics
EIT	eit	eit, EIT, EIT_Data
FluxMed Signals	fluxmed + signals	fluxmed_signals, FluxMed-Signals
FluxMed Parameters	fluxmed + parameters	fluxmed_parameters, FluxMed_Parameters
Servo-U	servo	servo_u, Servo-U, SERV0
Mindray	mindray	mindray, Mindray
Syringe	syringe	syringe, Syringe, syringe_pumps
Other	other	other, Other

3.3 Expected File Types

- **Philips Waves:** .parquet files
- **Philips Numerics:** Files containing “numerics” in the filename
- **EIT:** .asc files
- **FluxMed:** Files containing “signals” or “parameters” in the filename
- **Servo-U:** .sta files
- **Mindray:** .xml or .csv files
- **Syringe:** Files containing “syringe” in the filename
- **Other:** .csv or .parquet files (auto-discovers columns with datetime values)

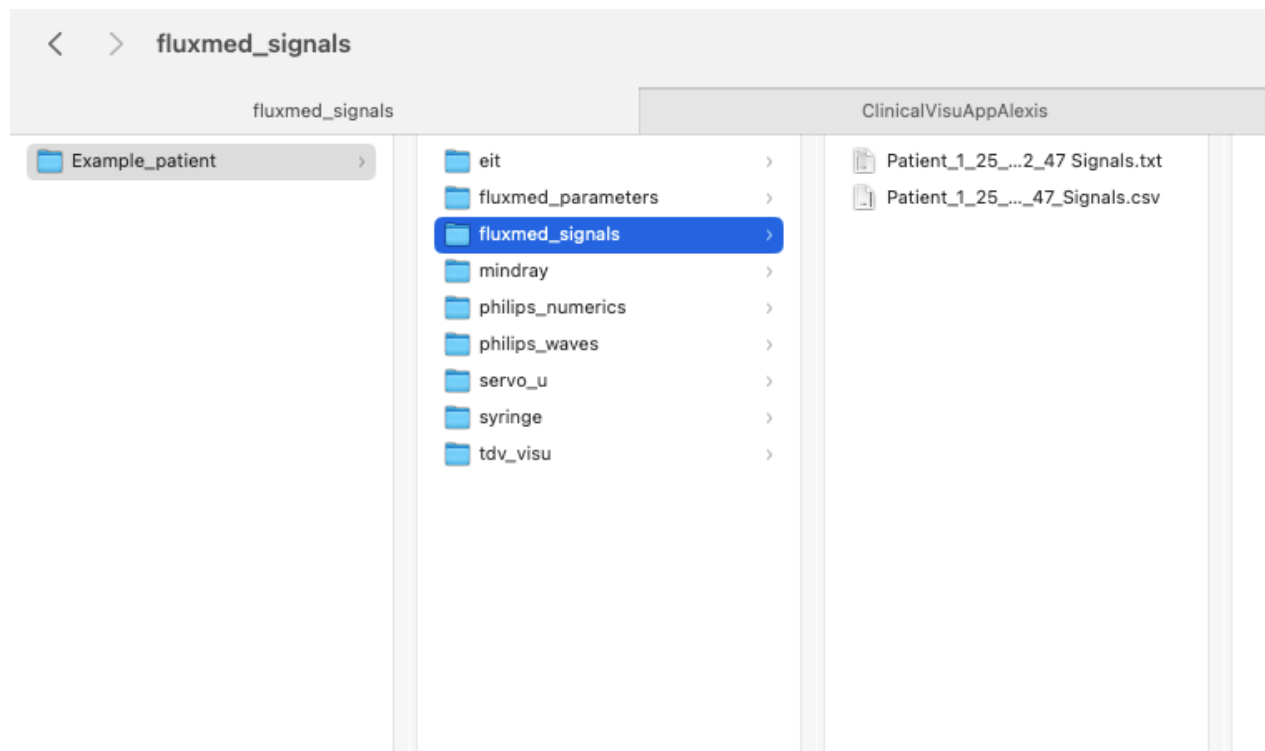


Figure 3: Patient folder structure example

4 Loading Database Options

Database options define **which data sources to enable** and **how signals should be displayed** (labels, units, colors, grouping).

4.1 Option 1: Default Visualization (Quick Start)

Click the green “**Default visualization (all sources)**” button. This automatically enables all 9 data sources with their default display settings. No configuration file is needed.

This is the recommended starting point for new users.

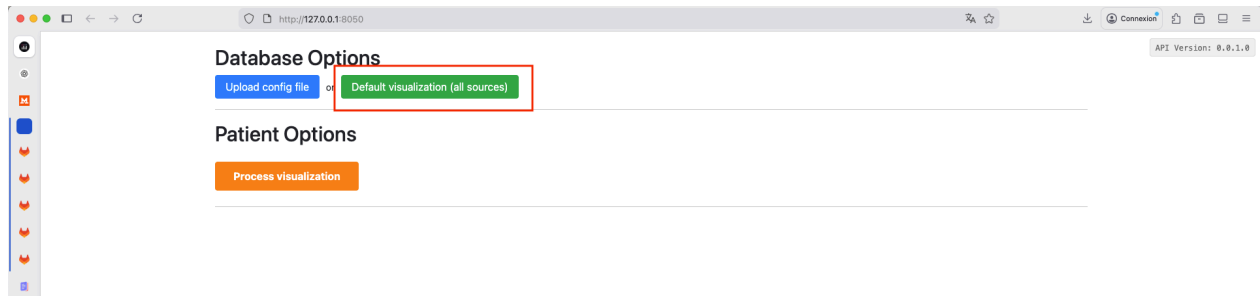


Figure 4: Default visualization button

4.2 Option 2: Custom Configuration File

Click the blue “**Upload config file**” button to load a custom `database_options.json` file. This gives you full control over which sources are enabled and how each signal is displayed.

See Section 9 for a detailed description of the configuration file format.

5 Configuring Patient Options

After loading database options, the **Patient Options** form appears. It is divided into two parts: global options and per-source options.

5.1 Global Options

These apply to all data sources:

Option	Description
Path to data (folder)	Full path to the patient's root data folder
Time start filter	Start of the time window to display (format: YYYY-MM-DD HH:MM:SS). Leave empty to use all available data.
Time end filter	End of the time window to display. Leave empty to use all available data.
Re-use data if already loaded once	When checked, reuses previously cached <code>.parquet</code> files from the <code>tdv_visu/</code> folder, significantly speeding up subsequent loads.

The screenshot shows the 'Patient Options' form. At the top, there's a 'Database Options' section with two buttons: 'Upload config file' (blue) and 'Default visualization (all sources)' (green). Below these is a status message: 'Using default visualization (all sources)'. The 'API Version: 0.0.1.0' is displayed in the top right corner. The main section is 'Patient Options', which includes a sub-section 'Global Patient Options'. This section contains a 'Path to data (folder)' input field, 'Time start filter' and 'Time end filter' input fields with a YYYY-MM-DD HH:MM:SS format, and a 'Re-use data if already loaded once' checkbox which is checked. Below this is the 'Specific Options' section, which shows two cards: 'Philios scope - waves' and 'EIT - PulmoVista'.

Figure 5: Global patient options

5.2 Per-Source Options

Below the global options, each enabled data source may have additional settings displayed in individual cards arranged in a two-column grid. Common per-source options include:

- **Time shift** (seconds): Adjust the time alignment of a source relative to others. Useful when devices were not perfectly synchronized.
- **Day**: Specify the recording date for sources that require it (e.g., EIT data).

Only data sources present in the loaded database options will show their configuration cards.

Specific Options

Philips scope - waves

Time shift

Philips scope - numerics

Time shift

Fluxmed - numerics

Time shift

Servo U

Time shift

Other (generic)

Time shift (seconds)

EIT - PulmoVista

Time shift

Day of EIT recording

Syringe

Time shift

Fluxmed - waves

Time shift

Mindray scope

Time shift

Process visualization

Figure 6: Per-source options

6 Processing the Visualization

Once you have configured the patient options, click the large orange “**Process visualization**” button to start generating the plots.

6.1 What Happens During Processing

1. **Validation:** The application verifies that all mandatory fields are filled in and that the data folder exists.
2. **Data Discovery:** For each enabled data source, the application scans the patient folder for matching subfolders and files.
3. **Data Loading:** Raw data files are parsed according to each source’s format.
4. **Formatting:** Signals are filtered, resampled, and converted using your database options (labels, units, time range).
5. **Caching:** Processed data is saved as `.parquet` files in the `tdv_visu/` subfolder for faster reloading next time.
6. **Plot Generation:** Interactive Plotly figures are created and displayed in the visualization area.

A success message appears when processing completes. If no data is found for a source, it is silently skipped.

API Version: 0.0.1.0

Database Options

Upload config file or Default: visualization (all sources)

Successfully loaded database_options.json

Patient Options

Global Patient Options

Path to data (folder)

Time start filter Time end filter

Re-use data if already loaded once ☒

Specific Options

Philips scope - waves Time shift 0	EIT - PulmoVista Time shift 0 Day of EIT recording YYYY-MM-DD HH:MM:SS
Philips scope - numerics Time shift 0	Syringe Time shift 0

Process visualization

Figure 7: Processing visualization

7 Interacting with Plots

7.1 Navigation Controls

Each plot provides a toolbar (top-right corner) with the following tools:

Tool	Action
Zoom	Click and drag to zoom into a rectangular region
Pan	Click and drag to move the view
Zoom In / Zoom Out	Incremental zoom buttons
Autoscale	Reset the view to fit all data
Reset Axes	Return to the original view
Download as PNG	Save the current plot view as an image

7.2 Zooming and Panning

- **Scroll wheel:** Zoom in/out on the x-axis.
- **Click and drag:** Select a region to zoom into.
- **Double-click:** Reset the axes to show all data.

7.3 Dynamic Resampling (FigureResampler)

For high-frequency signals (e.g., waveforms sampled at hundreds of Hz), the application uses **Plotly-Resampler** to dynamically load detail as you zoom in. When viewing a long time range, the plot shows a downsampled overview. As you zoom into a shorter time window, the full-resolution data is loaded automatically.

This keeps the interface responsive even with millions of data points.

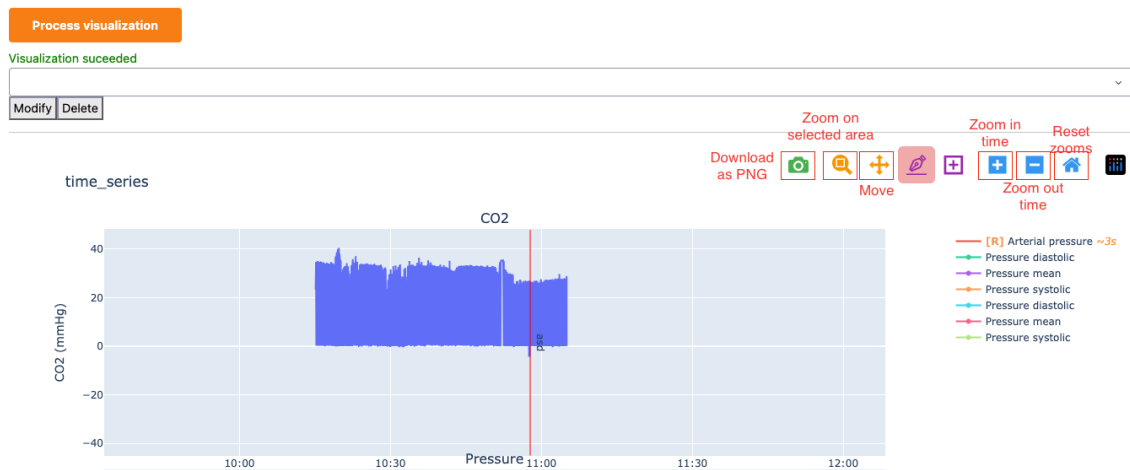


Figure 8: Interactive plot navigation

8 Annotations

The annotation system lets you mark events, time periods, or regions of interest directly on the plots. Annotations are saved to an `annotations.json` file in the `tdv_visu/` folder and persist across sessions.

8.1 Drawing Annotations

Use the Plotly drawing tools in each plot's toolbar:

- **Draw Line:** Click two points to draw a vertical line marking a specific event.
- **Draw Rectangle:** Click and drag to highlight a time region or value range.

Each annotations can be given a name and a color via the edit popup that appears after drawing.

8.2 Annotations Management

After processing, the **Annotation Controls** section becomes visible below the Process button:

- **Annotations Dropdown:** Lists all annotations across all figures, showing the annotations label.
- **Modify Button:** Opens the edit popup for the selected annotation (change name, color, or whether it spans all subplots).
- **Delete Button:** Removes the selected annotation.

8.3 Annotation Properties

Each annotation has the following properties:

- **Name:** A text label displayed on the annotation.
- **Color:** The line or fill color.
- **Global:** When enabled, the annotation spans all subplots in the figure (using paper y-coordinates).

8.4 Persistence

Annotations are automatically saved to `annotations.json` in the patient's `tdv_visu/` folder whenever you create, modify, or delete an annotation. They are reloaded when you re-process the same patient data.

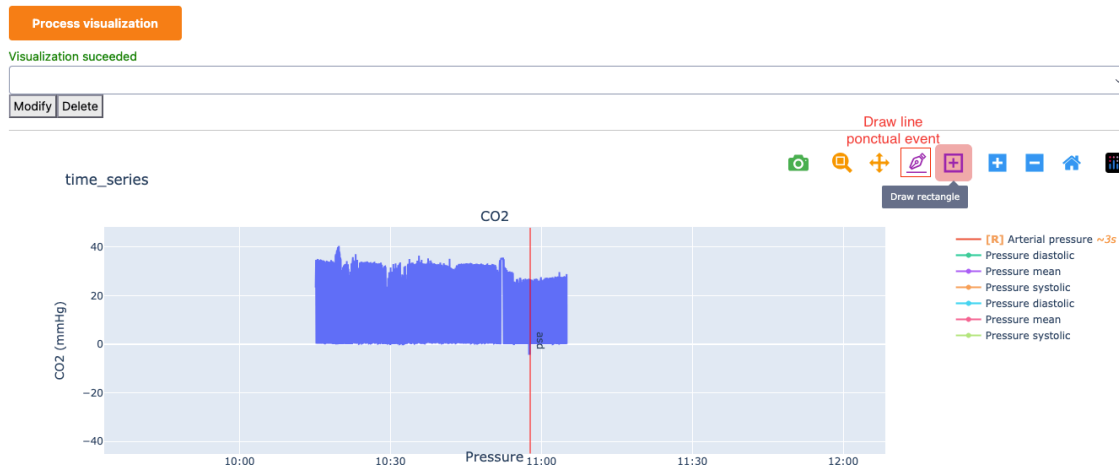


Figure 9: Annotations tools

9 Configuration File Reference

9.1 patient_options.json

This file defines patient-specific settings.

```
{
  "data_folder": "/path/to/patient/data",
  "datetime_start": "2024-10-08 10:00:00",
  "datetime_end": "2024-10-08 12:00:00",
  "quick_load": false,
  "philips_waves": {
    "time_shift": 20.0
  },
  "eit": {
    "day": "2024-10-08"
  }
}
```

Key	Type	Description
data_folder	string	Path to the patient's root data folder
datetime_start	string or null	Start of the time window (YYYY-MM-DD HH:MM:SS)
datetime_end	string or null	End of the time window
quick_load	boolean	Reuse cached parquet files
<source_name>	object	Per-source options (e.g., time_shift, day)

9.2 database_options.json

This file controls which data sources are active and how signals are displayed.

9.2.1 Top-Level Structure

```
{
  "global": {
    "grouped_fields": { "Group Name": ["signal1", "signal2"] }
  },
  "philips_waves": { ... },
  "philips_numerics": { ... },
  "eit": { ... }
}
```

Each data source key is optional – only include the sources you want to enable.

9.2.2 Per-Source Fields

Key	Description
field_display	Array of signal names to show (others are hidden)
data.label_correspondence	Map raw signal names to display labels
data.unit_conversion	Multiplication factors for unit conversion
data.unit_info	Display unit strings (e.g., “mmHg”, “cmH2O”)

Key	Description
<code>data.unit_range</code>	Fixed Y-axis ranges as <code>[min, max]</code>
<code>data.color</code>	Custom colors per signal
<code>data.priority</code>	Plot ordering priority (lower = higher on page)
<code>data.period_resampling</code>	Resampling period in seconds
<code>grouped_fields</code>	Group signals onto the same subplot
<code>loop</code>	Define PV-loop plots (e.g., <code>{"pv_loop": ["Pressure", "Volume"]}</code>)
<code>numerics.period_resampling</code>	Resampling for numeric parameters
<code>numerics.priority</code>	Plot priority for numerics

9.2.3 Global Fields

Key	Description
<code>global.grouped_fields</code>	Group signals from different data sources onto the same subplot

See `example/option_files/` in the source repository for complete example files.

10 Troubleshooting

10.1 Browser Does Not Open Automatically

If the browser does not open after launching the application, manually navigate to:

`http://127.0.0.1:8050`

Ensure no other application is using port 8050. If needed, close the terminal window which was opened in the app and restart the application.

10.2 No Data Found

If the visualization is empty or a data source shows no signals:

- Verify that the **data folder path** is correct and accessible.
- Check that subfolders follow the **naming conventions** (see Section 3).
- Ensure the subfolder contains files in the **expected format** for that data source.
- Check that the data source is **enabled** in your database options (or use “Default visualization” to enable all).

10.3 Slow Loading

Large datasets may take time to load on the first run. To speed up subsequent loads:

- Enable the “**Re-use data if already loaded once**” (`quick_load`) option. This uses the cached `.parquet` files in `tdv_visu/` instead of re-reading raw data files.

10.4 Time Alignment Issues

If signals from different sources appear misaligned in time:

- Use the **Time shift** option in the per-source settings to adjust alignment.
- Verify that the correct **day** or **date** is set for sources that require it (e.g., EIT).

10.5 Application Crashes or Errors

- Check the terminal window for error messages.
- Log files are available in the `logs/` directory (if running from source).
- Ensure the data files are not corrupted or truncated.

11 Appendix: Supported Data Sources

Source	Module Name	Keywords	File Types	Typical Signals
Philips Waves	<code>philips_waves</code>	<code>philips, waves</code>	<code>.parquet</code>	ART, PAP, CO2, respiratory pressure/volume
Philips Numerics	<code>philips_numerics</code>	<code>philips, numerics</code>	numerics in name	Heart rate, SpO2, FiO2, blood pressure
EIT	<code>eit</code>	<code>eit</code>	<code>.asc</code>	Global/local impedance, impedance percentages
FluxMed Signals	<code>fluxmed_signals</code>	<code>fluxmed, signals</code>	signals in name	Respiratory waveforms
FluxMed Parameters	<code>fluxmed_parameters</code>	<code>fluxmed, parameters</code>	parameters in name	Respiratory parameters
Servo-U	<code>servo_u</code>	<code>servo</code>	<code>.sta</code>	Ventilator waveforms and settings
Mindray	<code>mindray</code>	<code>mindray</code>	<code>.xml, .csv</code>	Monitor waveforms and parameters
Syringe	<code>syringe</code>	<code>syringe</code>	syringe in name	Infusion rates and volumes
Other	<code>other</code>	<code>other</code>	<code>.csv, .parquet</code>	Any time-series with datetime columns